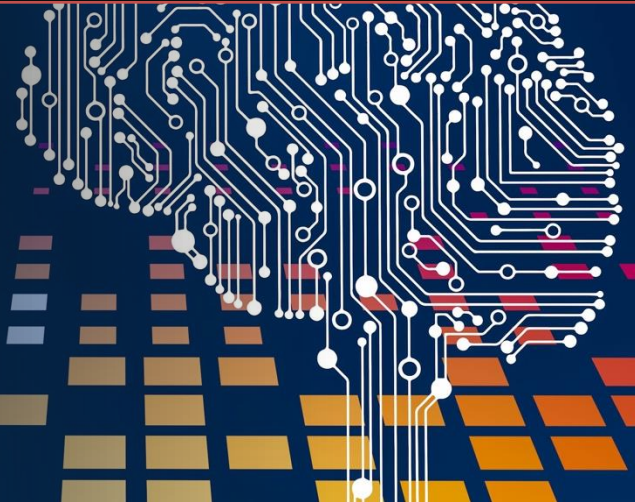


Technical Skill Training Program



GEMLABS

LEARN. INNOVATE. LEAD.



INTRODUCTION TO GENERATIVE AI



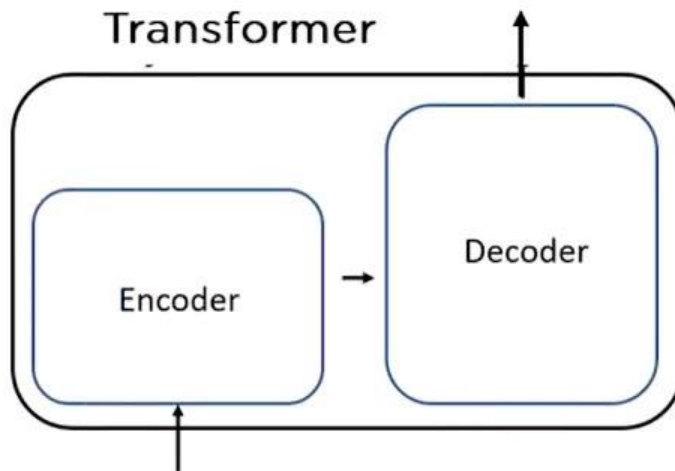
The Evolution: From RNNs to Transformers

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Unlike **RNNs**, **transformers** use an attention mechanism, allowing them to consider the entire sentence or paragraph simultaneously, rather than one word at a time

The Evolution: From RNNs to Transformers



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The Evolution: From RNNs to Transformers

Example:

As she said this, she looked down at her hands...

The Evolution: From RNNs to Transformers

Tokenization

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['As', 'she', 'said', 'this', ',', 'she', 'looked', 'down', 'at', 'her',
'hands', '...']

The Evolution: From RNNs to Transformers

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Tokenization

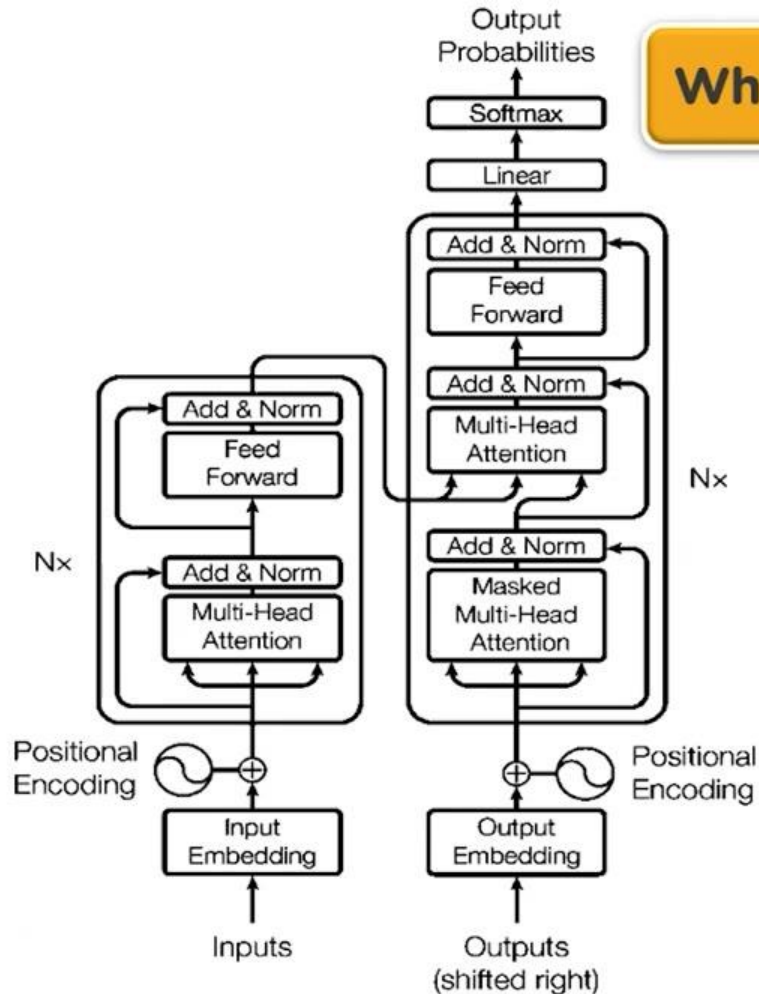
Encoding

Embeddings

Context Vector

Autoregressive
Generation

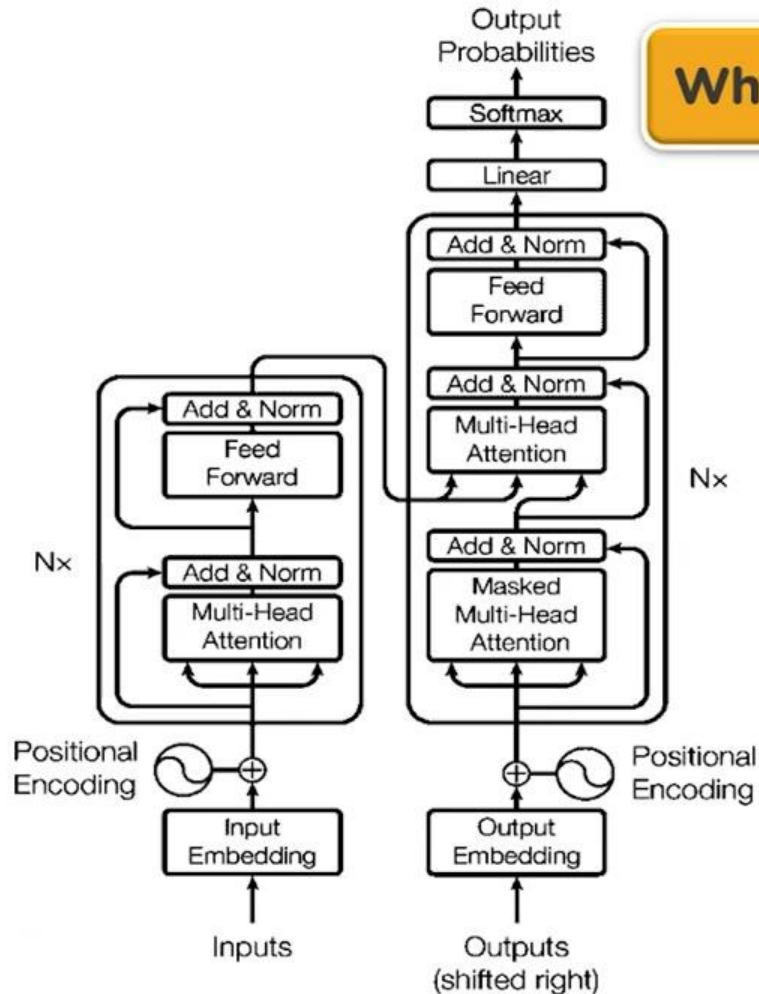
Why Transformers Can Predict Text



Why Predicting Text is Possible:

Redundancy in Language

Why Transformers Can Predict Text

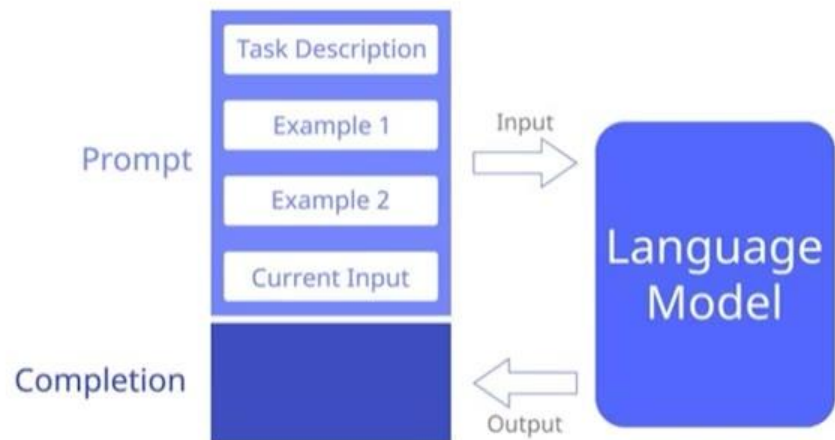


Why Predicting Text is Possible:

Redundancy in Language

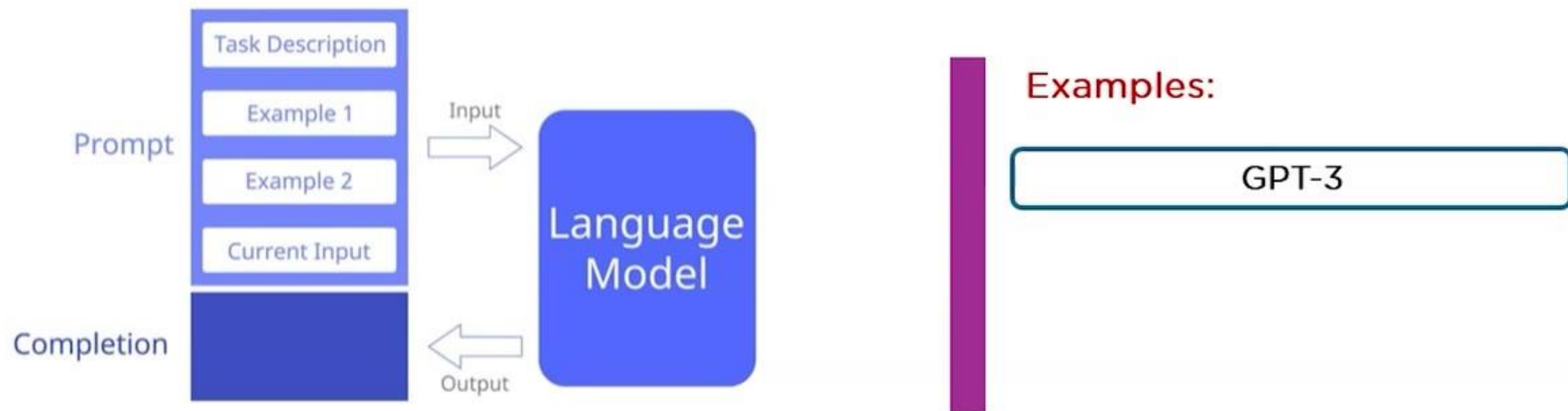
Entropy and Predictability

What is a Large Language Model?



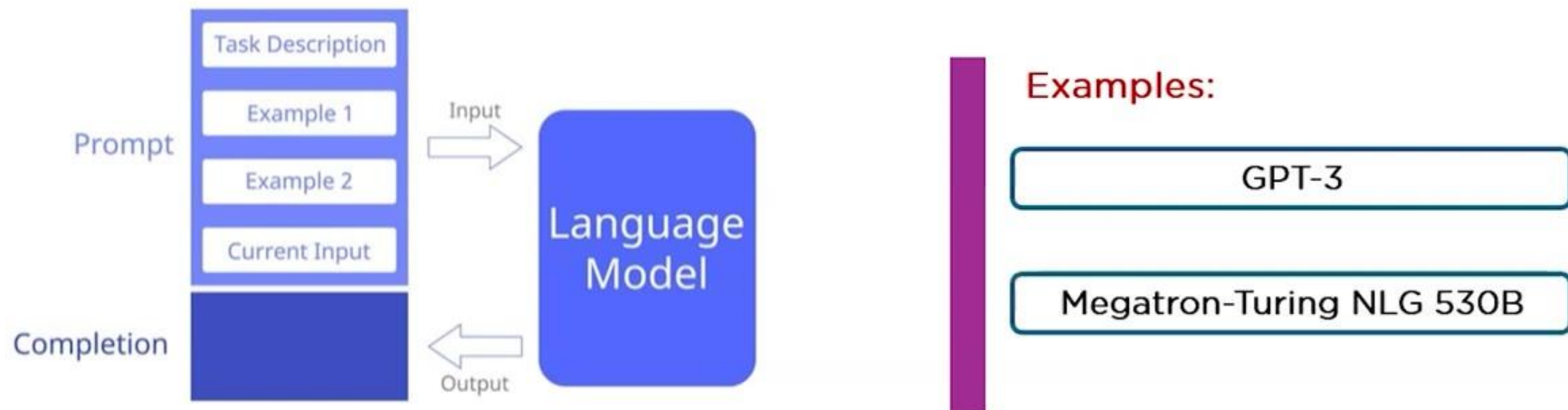
A **Large Language Model** is a type of artificial intelligence that uses deep learning and vast datasets to understand, summarize, generate, and predict new content. LLMs are a subset of **generative** AI, specifically designed to **create text-based content**

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Capabilities of LLMs

Generation



LLM

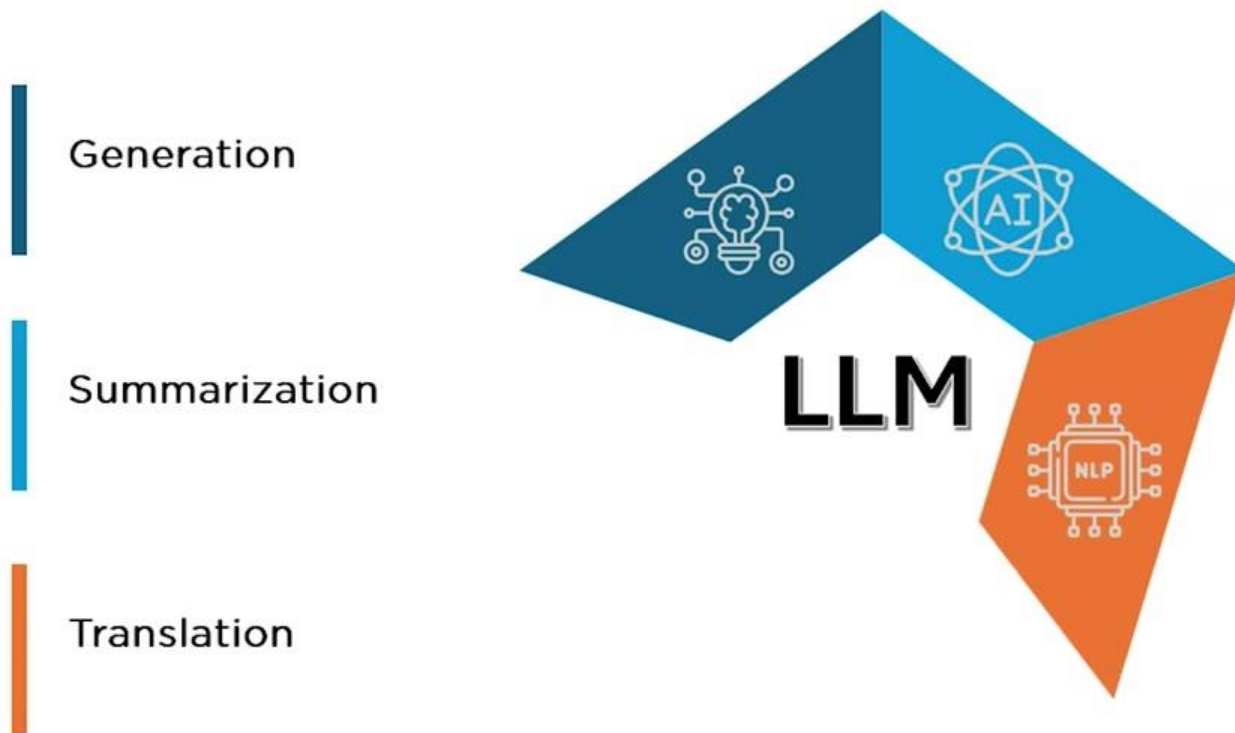
Capabilities of LLMs

Generation

Summarization



Capabilities of LLMs



Capabilities of LLMs



Capabilities of LLMs



Real-World Applications

Healthcare

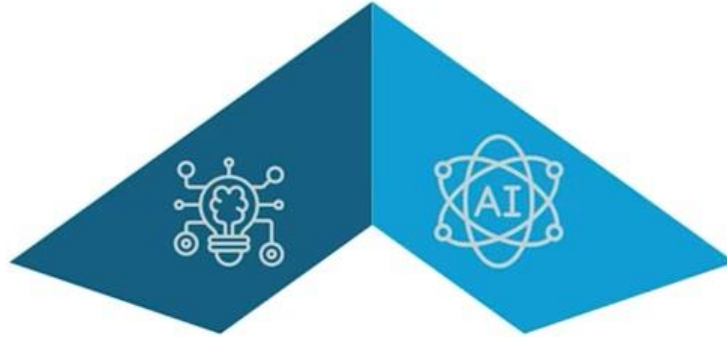


LLM

Real-World Applications

Healthcare

Retail

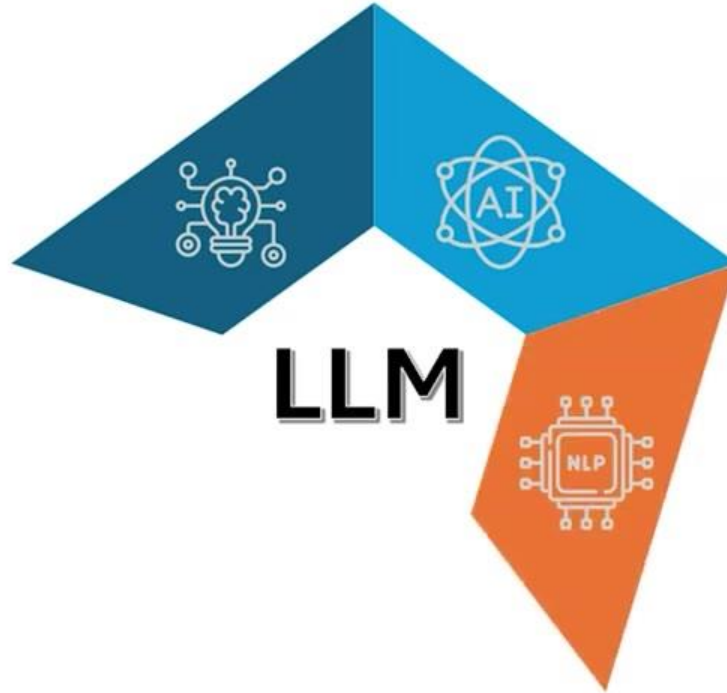


LLM

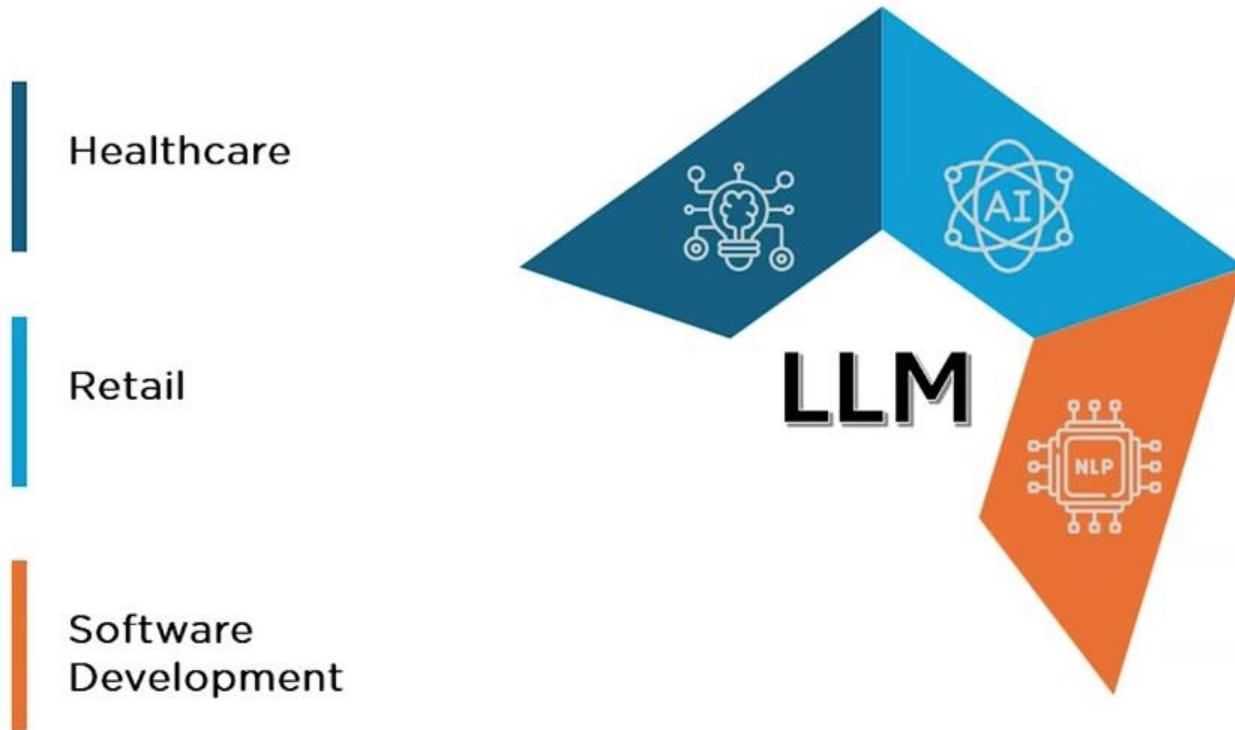
Real-World Applications

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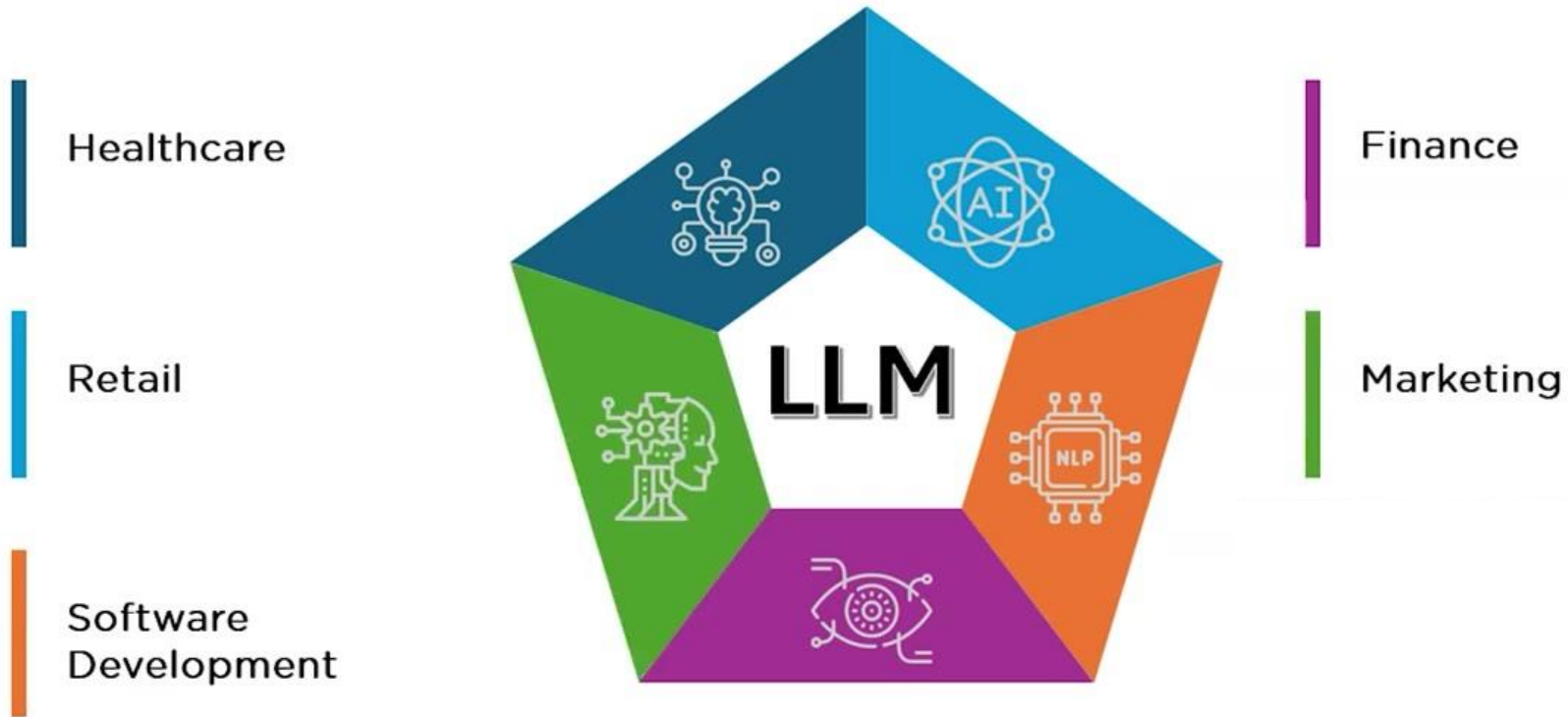
Real-World Applications



Real-World Applications



Real-World Applications



How Do Large Language Models Work?

LLMs are trained using **unsupervised learning**, which means they learn from large datasets without labeled data. They find patterns and relationships in the data, **enabling them to perform various tasks** like text generation, summarization, and translation



Learning



1

Zero-Shot
Learning

Learning



1

Zero-Shot
Learning



2

One-Shot
Learning

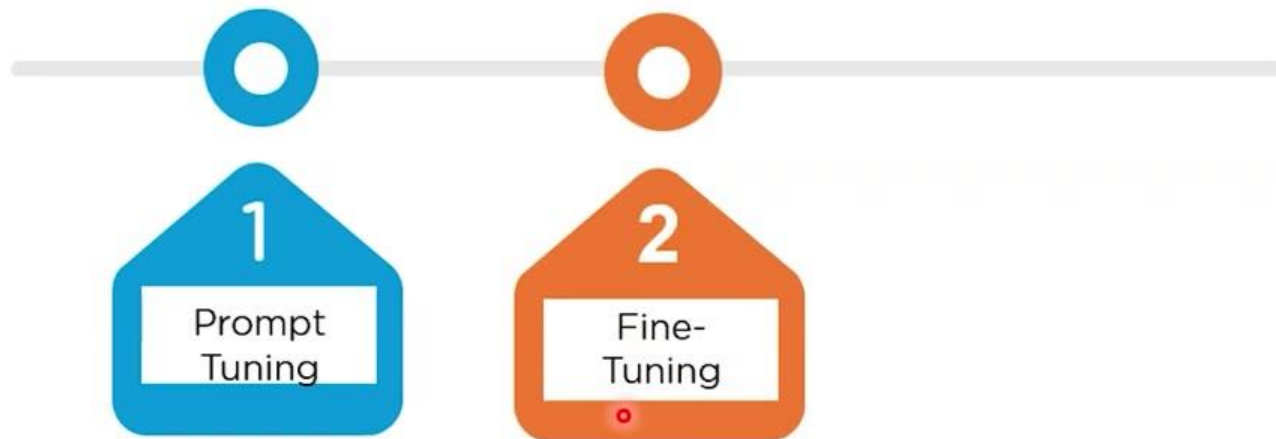
Learning



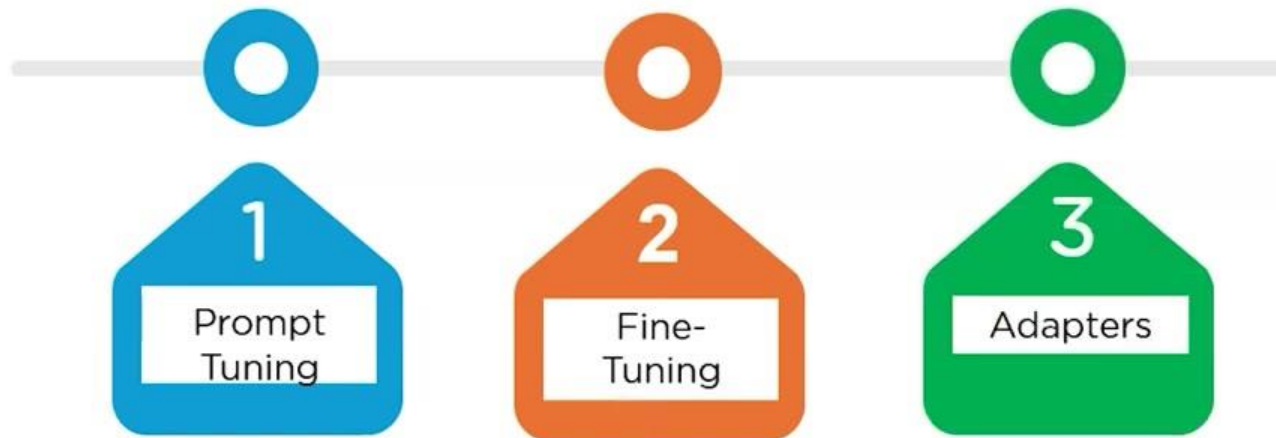
Customization Techniques



Customization Techniques



Customization Techniques



Types of Large Language Models



1

Encoder-
Only Models

Types of Large Language Models



Types of Large Language Models



Advantages and Challenges of Large Language Models

Advantages

Extensibility

Flexibility

Performance

Accuracy

Ease of Training

Efficiency

Advantages and Challenges of Large Language Models

Advantages

Extensibility

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Challenges

Development Costs

Operational Costs

Bias

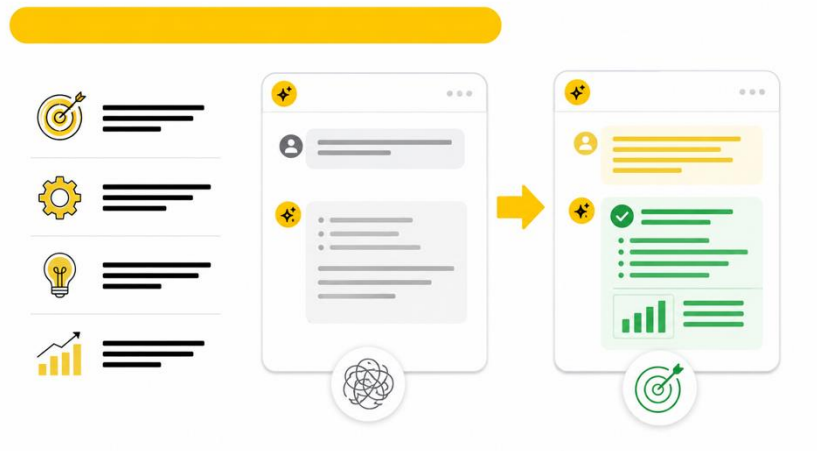
Ethical Concerns

Explainability

Hallucination

Prompt Engineering

A practical guide to designing prompts that produce clear, reliable, and useful AI outputs



Why Prompt Engineering Matters

A prompt is not just a question. It is the control layer that tells the model what role to take, what task to perform, what context to use, and what output format to produce.

1 More accurate outputs

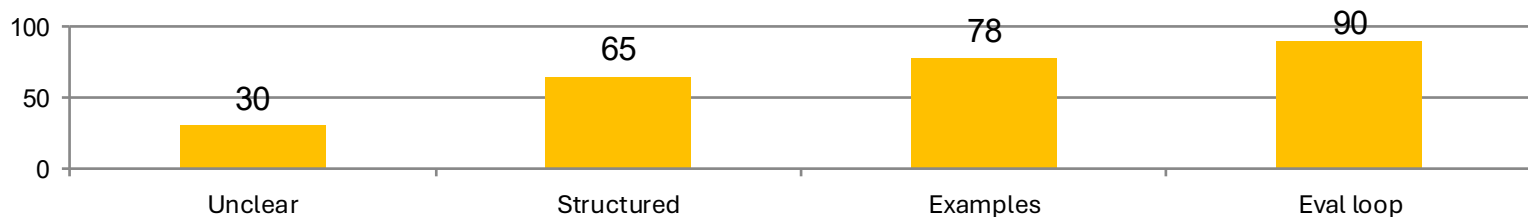
Precise instructions reduce ambiguity and help the model focus on the right objective.

2 Consistent format

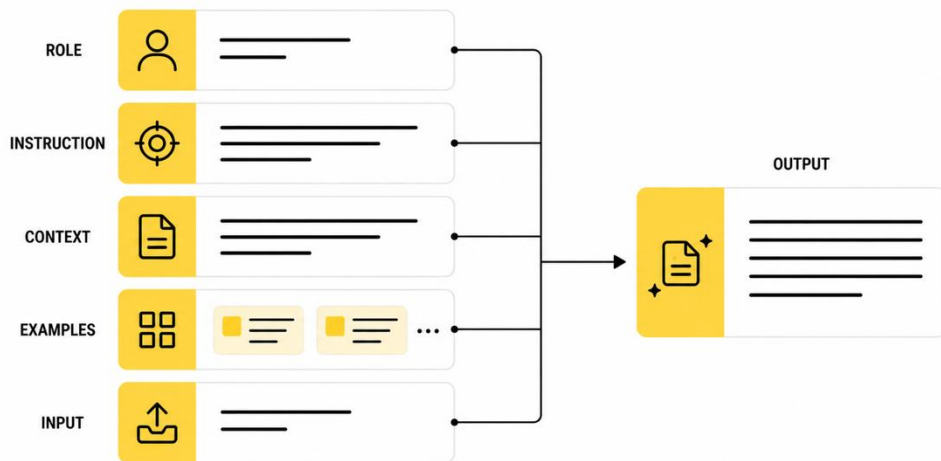
Structured output preferences make answers easier to read, grade, or automate.

3 Reusable workflows

Prompt patterns can be reused in chatbots, agents, RAG systems, and business workflows.



Anatomy of a Strong Prompt



Core components

- Role: who should the model act as?
- Instruction: what should it do?
- Output preference: format, length, style.
- Context: facts, constraints, audience.
- Examples: ideal input/output patterns.
- Input data: the real user request or data.

Formula: Role + Task + Context + Format + Examples + Input

Best Practices for Writing Prompts



Be clear and unambiguous

Write the task so that a new person could understand exactly what to do.



Be concise but specific

Long prompts can dilute instructions. Keep important details, remove noise.



Use consistent terminology

Avoid switching names for the same object or requirement inside one prompt.



Frame positively

Say what you want: “Use formal tone” instead of “Don’t be casual.”



Structure long prompts

Use headings or dividers such as ### Context, ### Task, ### Output.



Label data types

Clearly mark source text, user input, examples, and expected output.

Tip: Make the prompt easier for the model to parse, not just easier for humans to read.

Zero-Shot, One-Shot, Few-Shot

“Shot” means how many examples you include before the model answers.

0 Zero-shot

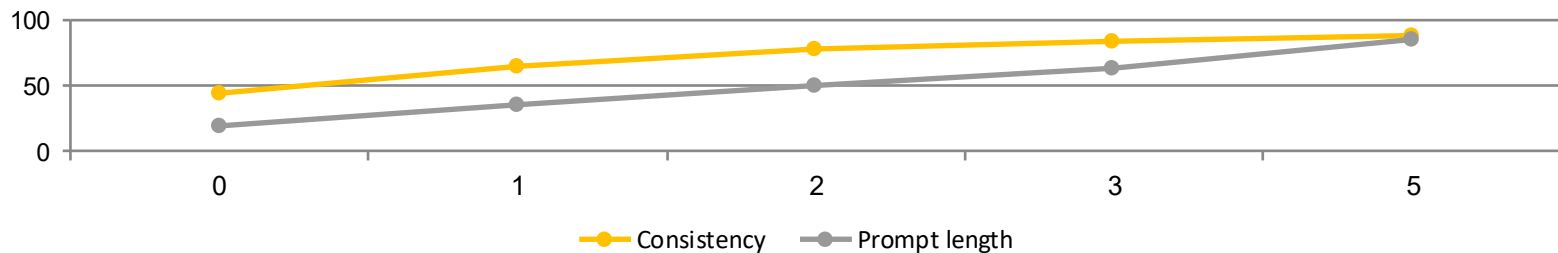
Instruction only. Fast and simple.
Best first test.

1 One-shot

One example teaches format, tone,
or logic.

5 Few-shot

2–5 examples improve pattern
consistency for classification and
structured output.



Match the Prompt to the Model

Different model types respond better to different prompt styles.

Model type	Best for	Prompt style	Avoid
SLM	Simple tasks	Short, direct	Long context
LLM	Writing, summary, ideation	Structured role + task	Vague tone requests
LMM	Text + image/screenshots	Label modalities clearly	Unlabeled visuals
Reasoning	Math, logic, planning	Ask for careful steps + checks	Rushed single-step tasks

Key idea: adapt the prompt to the model's strengths, limitations, and input capabilities.

Class Activity: Rewrite Prompts

Weak prompt

“Explain AI.”



Improved prompt

Role: You are an AI trainer.
Task: Explain AI to 3rd-year students.
Context: They know Python basics.
Output: 5 bullets + 1 simple example.
Tone: Clear and beginner friendly.

Reusable template

You are [role]. Your task is [instruction]. Use this context: [context]. Return the answer as [format]. Follow this style: [tone]. Here is the input: [input].

Practice: rewrite two weak prompts from your subject area using the template above.

Exercise 1: Plan exam preparation

STUDENT TASK

Student situation:

You have an internal test next week on Machine Learning. You need help preparing a practical 7-day study plan around daily college classes.

Write a clear prompt that includes:

- subject and exam date
- daily available study time
- topics to revise
- practice tasks for each day
- simple table format

Your prompt:

Rewrite this as a strong prompt: "Give me a study plan for ML exam."



Sample Answer 1: Plan exam preparation

MODEL PROMPT

Role

Context

Task

Format

Limit

Sample clear prompt:

Act as an academic mentor for a 3rd-year CSE student. I have an internal test on Machine Learning after 7 days. I can study 1.5 hours each evening after college. Create a 7-day study plan covering supervised learning, classification, regression, model evaluation, overfitting, and basic Python practice. Give the answer in a table with columns: Day, Topic, What to revise, Practice task, and Expected output. Keep it simple and realistic.

What makes it clear:

- daily time is given
- topics are listed
- role is defined
- table format is requested
- practical tasks are included

7
DAYS

Exercise 2: Understand lab code

STUDENT TASK

Student situation:

In your Python lab, you copied an LSTM text-classification program, but you do not understand the preprocessing and embedding layer.

Write a clear prompt that includes:

- programming language/tool
- your current knowledge level
- exact topics to explain
- line-by-line explanation request
- one small example

Your prompt:

Rewrite this as a strong prompt: “Explain this LSTM code.”



Sample Answer 2: Understand lab code

MODEL PROMPT

Role

Context

Task

Format

Limit

Sample clear prompt:

Act as a Python and Deep Learning lab instructor. I am a 3rd-year student and I understand basic Python, but I am new to NLP. Explain the preprocessing and embedding-layer part of an LSTM text-classification program in simple words. First explain tokenization, padding, and embedding. Then give a short line-by-line explanation using a small sentence example. Avoid advanced mathematics and keep the explanation suitable for a lab record.

What makes it clear:

- student level is clear
- scope is limited
- sequence is given
- example is requested
- math limit is added



Exercise 3: Prepare a lab record

STUDENT TASK

Student situation:

You completed a TensorFlow MNIST experiment and need to write the aim, algorithm, result, and viva questions for your lab record.

Write a clear prompt that includes:

- experiment name
- required lab-record sections
- beginner-friendly wording
- short result statement
- viva questions with answers

Your prompt:

Rewrite this as a strong prompt: “Make lab record for MNIST.”



Sample Answer 3: Prepare a lab record

MODEL PROMPT

Role

Context

Task

Format

Limit

Sample clear prompt:

Act as a lab assistant helping a 3rd-year engineering student. Prepare a lab record for the experiment “MNIST handwritten digit recognition using TensorFlow feed-forward neural network.” Include these sections only: Aim, Software Required, Algorithm, Model Design, Result, and 5 viva questions with answers. Use simple academic language. Keep the algorithm as numbered steps and write the result in 2 lines.

What makes it clear:

- experiment is specific
- sections are fixed
- style is stated
- numbered algorithm requested
- viva Q&A included



Exercise 4: Write a faculty email

STUDENT TASK

Student situation:

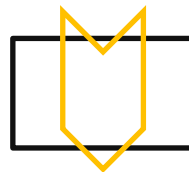
You were absent for one lab session due to a valid reason. You need to write a polite email requesting permission to submit the lab record late.

Write a clear prompt that includes:

- recipient and purpose
- polite tone
- reason in one line
- request for extension
- short subject line

Your prompt:

Rewrite this as a strong prompt: “Write email to mam for late submission.”



Sample Answer 4: Write a faculty email

MODEL PROMPT

Role

Context

Task

Format

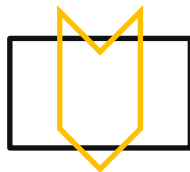
Limit

Sample clear prompt:

Act as a professional writing assistant. Draft a polite email from a 3rd-year student to the lab faculty requesting permission to submit the Python lab record two days late. Reason: I was absent due to a medical appointment. Include a short subject line, respectful greeting, 2 short body paragraphs, and a proper closing. Keep the tone sincere and not overly emotional.

What makes it clear:

- purpose is clear
- tone is specified
- reason is concise
- format is specified
- length is controlled



Exercise 5: Create mini-project presentation content

STUDENT TASK

Student situation:

Your team must present a mini project idea on AI for a 5-minute classroom presentation. You need slide-wise content.

Write a clear prompt that includes:

- project topic and audience
- presentation time limit
- number of slides
- slide-wise structure
- simple demo idea or dataset

Your prompt:

Rewrite this as a strong prompt: “Make PPT content for AI mini project.”



Sample Answer 5: Create mini-project presentation content

MODEL PROMPT

Role

Context

Task

Format

Limit

Sample clear prompt:

Act as a project guide for 3rd-year AI students. Create content for a 5-minute mini-project presentation on “Student feedback sentiment analysis using LSTM.” Prepare exactly 6 slides: Title, Problem Statement, Dataset, Methodology, Expected Output, and Future Scope. For each slide, give a title and 3 bullet points. Add one simple demo idea using 5 sample student comments. Keep the wording suitable for a classroom presentation.

What makes it clear:

- topic is specific
- time limit is given
- slide count is fixed
- slide titles are listed
- demo idea is requested



THANK

YOU !

